Huawei Access Node Configuration and Troubleshooting for Optical Transport Failure (SFP/XFP degradation, fiber cuts, Rx/Tx power loss) on BAS-ACC-37 at Basavanagudi POP

Node: BAS-ACC-37 | **Vendor:** Huawei | **Severity:** Warning | **Alarm Code:** ALARM_7523

Description

The Huawei Access Node (BAS-ACC-37) at Basavanagudi POP is a critical component of the telecommunications network, responsible for transmitting and receiving optical signals over the optical transport network. The failure of the SFP/XFP module or a fiber cut in the optical transport network can cause a degradation in the Rx/Tx power, leading to a warning alarm (ALARM_7523). This is due to the fact that the Huawei Access Node relies on the SFP/XFP module's optical power to transmit and receive data. The SFP/XFP module is responsible for converting the electrical signal from the line card to an optical signal that can be transmitted over the fiber optic cable. If the SFP/XFP module is degraded or a fiber cut occurs, the optical signal is weakened, resulting in a degradation of the Rx/Tx power. This can cause the Huawei Access Node to generate a warning alarm (ALARM_7523), indicating a potential issue with the optical transport network.

Pre-requisite Checks

- Verify that the Basavanagudi POP is accessible and physically secure.

- Check the redundant link status to ensure that it is operational and can take over in case of a failure.

- Gather all necessary equipment and spare parts, including the Huawei Access Node's configuration files and the SFP/XFP module.

Potential Root Causes

Cause	Probability
Degraded SFP/XFP Module	High
Fiber Cut in Optical Transport Network	Medium
Rx/Tx Power Loss due to Optical Signal Attenuation	Low

Diagnostic Commands & Expected Outputs

```
Command: display optical-transport interface status, Expected Output: Optical transport interface status: Normal, Description: This command is used to display the status of the optical transport interface, including the SFP/XFP module's optical power and the fiber optic cable's link status.
```

```
Command: display optical-transport interface detail, Expected Output: Optical transport interface detail: SFP/XFP module degraded, Description: This command is used to display detailed information about the optical transport interface, including the SFP/XFP module's status and the fiber optic cable's link status.
```

Resolution Steps

Step 1: Step: Replace the Degraded SFP/XFP Module, Command: replace optical-transport interface sfp/xfp-module, Description: This step involves replacing the degraded SFP/XFP module with a new one. The Huawei Access Node will automatically detect the new module and configure it accordingly.

Step 2: Step: Check and Replace the Fiber Optic Cable, Command: replace optical-transport interface fiber-optic-cable, Description: This step involves checking the fiber optic cable for any damage or cuts. If a cut is found, replace the cable with a new one. The Huawei Access Node will automatically detect the new cable and configure it accordingly.

Post-Check Verification

```
Command: display optical-transport interface status, Expected Output: Optical transport interface status: Normal, Description: This command is used to verify that the optical transport interface is functioning normally after the resolution steps have been completed.

Command: display optical-transport interface detail, Expected Output: Optical transport interface detail: SFP/XFP module normal, Description: This command is used to verify that the SFP/XFP module is functioning normally after the resolution steps have been completed.
```

Roll-back Procedures

- Step: Restore the Original SFP/XFP Module, Command: restore optical-transport interface sfp/xfp-module original, Description: This step involves restoring the original SFP/XFP module that was replaced during the resolution steps. This will ensure that the Huawei Access Node returns to its original configuration.

- Step: Restore the Original Fiber Optic Cable, Command: restore optical-transport interface fiber-optic-cable original, Description: This step involves restoring the original fiber optic cable that was replaced during the resolution steps. This will ensure that the Huawei Access Node returns to its original configuration.